

A Deep Learning Framework for Recognition and Classification of Diabetic Retinopathy Severity

**Dr. Ranga Swamy Sirisati¹, Dr. Gunji Sreenivasulu², Dr. Pradeep Venuthurumilli³,
Dr. A.C. Priya Ranjani⁴, Dr. Kanusu Srinivasa Rao⁵**

¹Associate Professor, Department of Computer Science & Engineering, Vignan's Institute of Management and Technology for Women, Kondapur, Ghatkesar.

²Assistant Professor, Department of Computer Science & Engineering, Madanapalle Institute of Technology & Science, Madanapalle, AP, India.

³Associate Professor, Department of CSE-Data Science, Malla Reddy Engineering College for Women, Maisammaguda, Telangana 500100

⁴Associate Professor, Dept. of CSE, Vijaya Institute of Technology for Women, Vijayawada.

⁵Assistant Professor, Department of Computer Science and Technology, Yogi Vemana University, Kadapa-516005

Email: ¹sirisatiranga@gmail.com, ²gsr1251@gmail.com, ³pradeepvenuthuru@gmail.com, ⁴acpranjani@gmail.com, ⁵kanususrinivas@gmail.com

Received 29/01/2024;

Accepted 23/02/2024

Abstract

Diabetes is an infection that improves when the human body cannot properly utilize insulin or the pancreas fails to release enough. When diabetes worsens, the condition increasingly affects the cardiovascular system. If DR is not diagnosed at the earliest possible stage, it can result in partial or complete vision loss. Retinal lesions related to the infection are used to determine the several stages and the severity of the condition. Retinal images taken using a fundus camera with a motorized camera on the rear provide useful details of the nature, outcome, and stage of effects on the eye. These kinds of pictures aid ophthalmologists in evaluating patients to plan various administration strategies and make progress monitoring more effective. If DR is not treated, it could eventually become the primary cause of vision loss and visual impairment. Those with diabetes have a rise in blood glucose levels and retinal damage. Blindness or vision loss results from this. A wide range of conditions, including cataracts, glaucoma, DR, and age-related macular degeneration, are major causes of blindness and visual impairment in people. DR is the colloquial name for the blinding diabetic eye condition. As a result, the blood cells in the retina are injured, which could cause blindness. Retinopathy is a disease that damages the blood vessels that supply the retina. This disorder can cause distorted vision, eye hemorrhage, and total vision loss. This disease impairs the visual nerve by obstructing blood flow.

Keywords: Diabetic Retinopathy, Machine Learning Algorithms, Deep Learning Algorithms, Deep Neural Network Classifier.

1. INTRODUCTION:

Retinal Image of the Human Eye

The retina receives an image of the visible world via the eye's optics, which functions similarly to the camera's film. When light hits the retina, chemical and electrical events begin

that eventually cause nerve-driving forces. Via the filaments of the optic nerve, these are transmitted to various visual foci of the cerebral cortex. Neurotransmitters connect the retina's several layers of neurons, which are arranged in a layered pattern. The first neurons that are especially sensitive to light are the photoreceptor cells.

Different retinal neurons process the rod and cone neurons' signals. Retinal ganglion cells, whose axons constitute the optic nerve, can move in a certain way thanks to the yield. A few retinal' encoding and light processing are important visual observation elements. The retina is typically red due to the abundant blood flow, and any changes in shading indicate retinal disease. The optic disc is often rounded and snug as a violin, and it is brighter than any portion of the retinal image [6]. Additionally, it serves as the point of entry and exit for nerves that pass through the retina on their way to and from the brain. The key components of the human eye responsible for vision are described and articulated here. Figure depicts an eye's retinal picture.

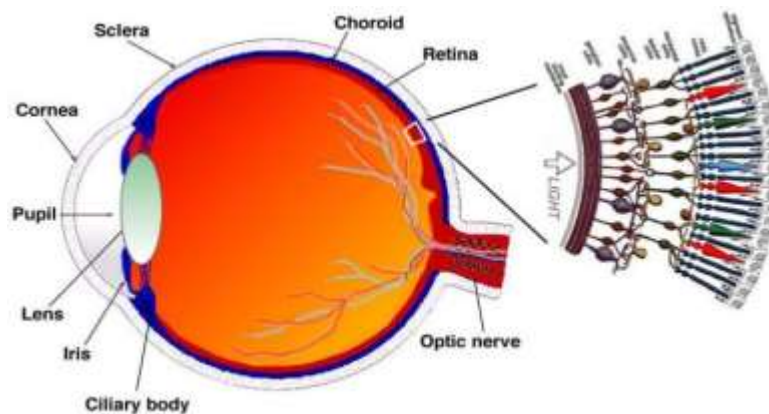


Figure1 : Retinal image of the eye

Iris: The eye's iris is its vibrant region, generally colored green, blue, and brown. In response to how bright the surrounding light is, it contracts and expands while opening and closing the pupil

Pupil: The iris's central, circular opening. The brightness of the iris muscle determines its size.

Retina: The retina is the eye's innermost, lightly tactile layer. It has rods and cones for detecting color and light.

Macula: The tiny area of the retina that is generally in focus in the back of the eye is known as the macula. It is the area where a person can focus on their blindness. The focus area of the macula is the fovea. It's an oval-shaped, intensely colored yellow patch near the retina's central region [8]. The retina's tiny, very sensitive region governs each point's ability to see in the center of the field. The fovea is pushed with cone cells. Due to the high density of light-sensitive cells in the fovea, it is this region that is in charge of providing the clearest vision.

Sclera: The screechy, stringy, protective outer covering and the eye's outermost bulk are collectively called the sclera. It preserves the condition of the eye and protects the internal organs.

Choroid: The vascular layer of the eye's choroid presents joining tissue between the sclera and retina. It nourishes the retina with blood. In unusual circumstances, veins form in the choroid zone and injure the macula.

Vitreous Humour: The area between the retina and the eye's lens is filled with a translucent, gloomy fluid with a jelly consistency. As we age, it transforms from a gel to a fluid and gradually compresses, isolating from the retina.

Crystalline lens: This is responsible for centering the images across the retina and is just behind the cornea.

Ciliary body: The middle eye layer is called the muscle or ciliary body, which controls and manages the flow of liquid intelligence.

Optic nerves: The optic nerve delivers the forces that give people their ability to see by connecting the eye and the brain. It is a region of the retina that is also known as the "optic disc."

Optic nerve head: It causes nerves to shoot up. This method introduces no tactile cells, creating a blind side. Every eye fills in the information that is absent since each eye covers the blind side of the other eye.

Abnormalities of Human Eye

Eye-related anomalies can be classified into two main forms. The first category includes eye defects such as conjunctivitis, glaucoma, and cataracts. The second group is organized in a certain way [9]. Various diseases are associated with life, such as diabetes, hypertension, and arteriosclerosis. DR is the term used to describe a condition where the retina is affected by diabetes and may eventually cause vision loss if not orderly treated. Ophthalmologists agree that the best treatment for this illness is early detection and treatment. Below figure shows the abnormal DR image.

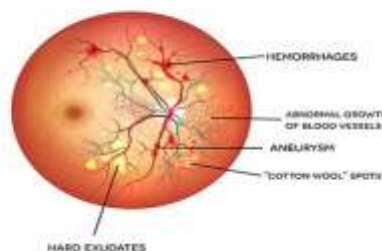


Figure2: Abnormal Image of DR

Diabetic Retinopathy

Diabetes complications can affect the retina of the human eye, called DR, which can ultimately result in blindness. Figure 1.3 shows the influenced retinopathy eye image. Up to 80% of diabetics condition for an extended period or longer DR, a visible symptom of a systemic disease [10]. Despite these unsettling revelations, research shows that at least 90% of the eyes were treated properly and courteously, and the number of these new cases might be reduced. The likelihood of improving DR increases with the length of a man's diabetes. Microvascular retinal alterations have the side consequence of DR. The thickening of the storm cellular film caused by hyperglycemia resulted in inadequate vascular segmentation. These damages cause retinal veins to become more pliable. It has a long-term diabetic effect.

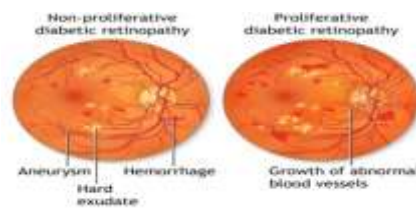


Figure3: Image for both NPDR and PDR

Diabetes's effects on the eye result in DR. It can damage the retina's tiny veins, which could in eyesight loss shown in

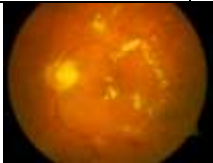
		
Normal Stage 0	Early Stage 1	Moderate Stage 2
		
Severe Stage 3	PDR Stage 4	

Figure4: Image for both NPDR and PDR

When delayed diabetes affects the retina, it is known as DR, which can lead to visual impairment. The best method of treatment is early detection and intervention. Figure 1.6 shows the image for exudates. These are classified into two classes, which are

- Red lesions (Haemorrhages/ Micro Aneurysms)
- Bright lesions (Exudates)

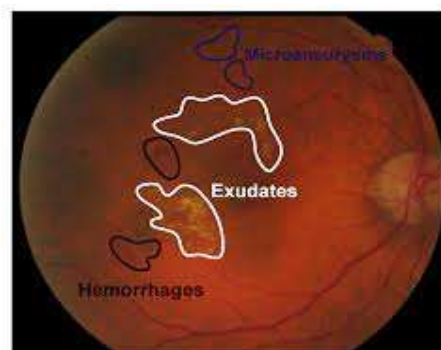


Figure5: A model image of exudates

LITERATURE REVIEW:

Different Types of Datasets

Messidor Database : MESSIDOR came from the Modified National Institute of Standards and Technology database. The MESSIDOR database was created to promote a focus on discoveries of DR with computer assistance. This database's information can be used for exploration and educational purposes with no out-of-pocket costs. Using a color video 3CCD camera and a Topcon TRC NW6 non-mydratic microscope with a 45-angle viewing angle, three ophthalmology clinics amassed the 1200 fundus color numerical photographs of the back shafts in the MESSIDOR system [19]. Eight bits per color plane were used to capture the photos at 2240 x 1488, or 2304 x 1536, or 1440 x 960 pixels. Eight hundred pictures were collected, 32400 without enlargement and 800 with understudy expansion. Three sets, one for each segmentation performed by an ophthalmologist, total of 1200 photos.

DIARETDB Database: This is addressed by DIARETDB0 and DIARETDB1. This is an open database for evaluating the recognition of DR in challenging images. The main goal of the outline has been to develop a database and a testing procedure that can be utilized to compare various DR detection techniques in detail. It is possible to compare the outcomes of various methods that employ this database and the established testing technique. The databases can be downloaded and used without charge for exploratory study. The copyrights to DIARETDB0 and Constitutionality were acquired by Tomi Kauppi and his team in 2006 and 2007, respectively.

Digital Image Processing

A diverse range of applications is incorporated into digital image processing. The field of medical image processing involves the computation of several picture types. For instance, thermal, fundus, and x-ray imaging. Utilizing specific medical tools, well-established procedures, and cutting-edge technologies, vital information about the human organ is collected and gleaned from the images. These methods are used to produce images that depict the internal organization of human organs. These images, which are recorded in energy format and transformed into signals, are produced by high-speed equipment. The diagnosis of any ailment in a human organ follows the processing of these signals. Image processing is used for enhancement, scrutiny, and highlighting. These systems have been employed for fundus imaging and research measurement. Several operations can be carried out on a computerized image depending on the prerequisites. Automated disease detection is crucial for accurate and timely disease prognosis.

Digital imaging

A two-dimensional signal is all that a digital image. The numerical capacity $f(a,b)$ where a and b are the two coordinates on a level plane and vertically distinguishes it. The pixel value of a picture is then used to estimate at any moment. A pixel, also known as a picture element, is the least elastic part of a photograph displayed on a screen since it is a physical point in a raster image or the smallest addressable component in an all-focus addressable presentation device in digital imaging. Only a collection of pixels makes up an advanced digital Picture.

Image Pre-Processing

Images are commonly pre-processed by removing low-frequency background noise, modifying the force of each particle picture, avoiding reflections, and obscuring portions of the images. Using computation, picture pre-processing is a strategy for improving data and earlier image processing.

Image pre-processing and optical analysis can enhance an optical assessment's reliability. Many filter processes that boost or reduce certain picture points of regions allow a

less labor-intensive or speedier assessment. Pre-processing is an essential first step in image inspection because of the wide range of determination, picture group, testing models, and lighting systems utilized during acquisition. Pre-processing aims to alter the image data to eliminate undesired changes or enhance certain image details crucial for subsequent processing.

2. METHODOLOGY:

Deep Learning Models for Detection and Classification

The learning strategy for streamlining the image features heavily relies on activation. The deep learning network's main benefit is that it improves accuracy in disease detection. It automatically determines how resilient the data is to common fluctuations. The same neural network-based approach can address various applications and data types. Massively parallel processes on GPUs can handle enormous volumes of data. Also, it functions better when there is a lot of data. Compared to other strategies, it requires a significant amount of data, which is one of its downsides, and getting results is not simple. To extract spatial features from images CNNs have been used.

Compared to manual diagnoses, automatic models for the detection of DR are both time and financially advantageous and highly effective. A manual diagnostic requires more work and is more prone to mistakes than automated techniques. ML approaches that incorporate a layer of hierarchical nonlinear handling stages for classifying patterns and unsupervised learning features are together known as deep learning (DL). Medical image analytics using DL include detection, segmentation, registration, classification, and picture retrieval. Recently, the DR diagnostic process has made extensive use of DL. When numerous diverse heterogeneous sources are used, it can successfully learn the features of the input data. It uses several DL-based techniques, including auto-encoders, CNN, limited Boltzmann machines, and sparse coding.

Motivation

There are several challenges associated with DR diagnosis and management, including it is difficult to diagnose in its initial phases as patients may not experience any symptoms. Dilated eye exams, which involve examining the retina after the pupil is dilated, are currently the gold standard for DR diagnosis. However, this procedure might take some time, and there is a global scarcity of eye care professionals. Effective and affordable screening techniques are required to identify DR in diabetic individuals. Telemedicine and AI-based screening methods are being developed to address this challenge. Current treatments for DR include laser surgery, injections, and vitrectomy. However, these treatments can be invasive and costly, and there is a need for more effective and less invasive treatments. Research in DR is motivated by the need to address these challenges and improve the diagnosis, screening, and disease treatment. Some of the areas of active research include: Researchers are developing AI-based methods for DR screening that can analyze retinal images and detect signs of DR with high accuracy. Researchers are investigating new biomarkers and imaging techniques that can spot the early signs of DR before it causes irreversible damage to the retina. Researchers are exploring new treatments for DR that are less invasive and can be administered without the need for surgery and investigating ways to personalize treatment for DR based on individual patient characteristics, such as genetics and biomarkers. Overall, research in DR aims to improve the diagnosis, screening, and treatment of the disease, and ultimately, prevent vision loss and blindness in diabetic patients.

Research Objectives

1. **Regional Segmentation:** A modified ResU-net approach is employed for regional segmentation of retinal fundus images. The resulting segmented image is then fed into the HSADMGR Net, a feature extraction deep learning model. The Metaheuristic method is used to optimize the model's parameters. Multi-Armed Bandit Ground Water Flow Algorithm (MABGFA) to ensure optimal performance of severity classification.
2. **Severity Grade Classification:** The proposed model uses VGG16 and ResNet to extract two sets of features (F1 and F2) from enhanced and segmented images, respectively. Utilizing weights that have been improved by the Modified Mating Probability-based Water Strider Algorithm (MMP-WSA), the features are combined. Then, utilizing the proposed MMP-WSA, the Optimized Deep Neuro-Fuzzy Classifier (ODNFC) is used for multi-class severity classification with hyper-parameter optimization. This approach improves the precise detection of abnormalities and better classification results.
3. **DL-DRDC Mechanism:** The suggested DL-DRDC mechanism employs the modified Efficient Net strategy to extract feature vectors from pre-processed retinal fundus images. A DNN classifier is then used to classify the feature vectors to get precise class labels. Performance evaluation is conducted by comparing the results with traditional models, considering various functionality factors.

3. RESULTS ANALYSIS

The experiments are run on Windows 8.1 using Python and the standard datasets. The following table represents the results of a performance comparison of existing classification algorithms and the proposed DRPNN in terms of precision, recall, and F-score metrics.

The table 1, summarizes the performance of various classification algorithms on the MESSIDOR dataset, focusing on metrics such as precision, recall, and F1-score. The proposed technique consistently achieves the highest values across all metrics, indicating its superior performance in accurately classifying data points in the MESSIDOR dataset.

Table -1: MESSIDOR Dataset Performance Results

Dataset	Classifiers	Precision	Recall	F-Score
MESSIDOR	GBR	78	74	76
	SVM	79	71	73
	LR	78	75	76
	DT	81	76	78
	k-NN	84	79	81
	RF	87	81	83
	PROPOSED	91	86	88

Similar to Table 1, this table presents the performance of different classifiers on the IDRID dataset. Again, the proposed technique stands out with the best scores for precision, recall, and F1-score, demonstrating its effectiveness in handling data from the IDRID domain.

Table -2: MESSIDOR Dataset Performance Results

Dataset	Classifiers	Precision	Recall	F-Score
IDRID	GBR	76	73	72

	SVM	75	71	72
	LR	78	75	73
	DT	82	76	79
	k-NN	83	77	80
	RF	86	81	82
	PROPOSED	90	85	87

Following the same structure, this table showcases the classification results on the APTOS dataset. Once again, the proposed technique outperforms other methods in terms of precision, recall, and F1-score, solidifying its overall efficacy across diverse datasets.

Table -3: APTOS Dataset Performance Results

Dataset	Classifiers	Precision	Recall	F-Score
APTOS	GBR	78	74	72
	SVM	79	75	73
	LR	80	72	75
	DT	84	78	79
	k-NN	86	79	82
	RF	89	83	80
	PROPOSED	91	86	88

This Figure-1 visually compares the accuracy of different classifiers across all three datasets (MESSIDOR, IDRID, and APTOS). The proposed technique consistently maintains the highest accuracy level, highlighting its ability to make accurate predictions regardless of the dataset.

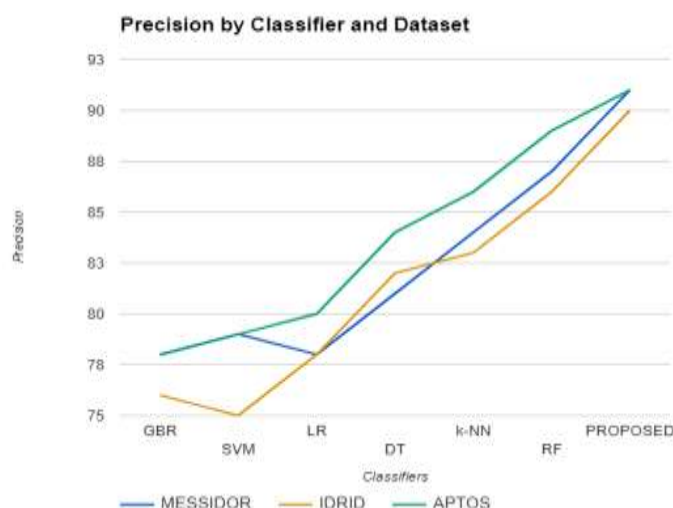


Figure 6: Accuracy Comparison

Similar to Figure 1, this Figure-2 focuses on recall, showcasing how well each classifier identifies true positives within each dataset. The proposed technique exhibits the best recall performance, indicating its strength in capturing relevant data points.

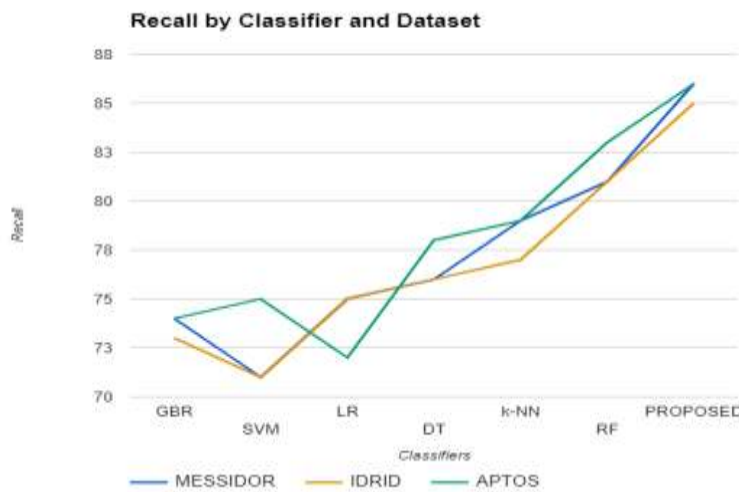


Figure 7: Recall Comparison

This Figure-3 combines precision and recall into a single metric, the F1-score, providing a more balanced view of classifier performance. Once again, the proposed technique achieves the highest F1-scores across all datasets, signifying its overall effectiveness in both precision and recall.

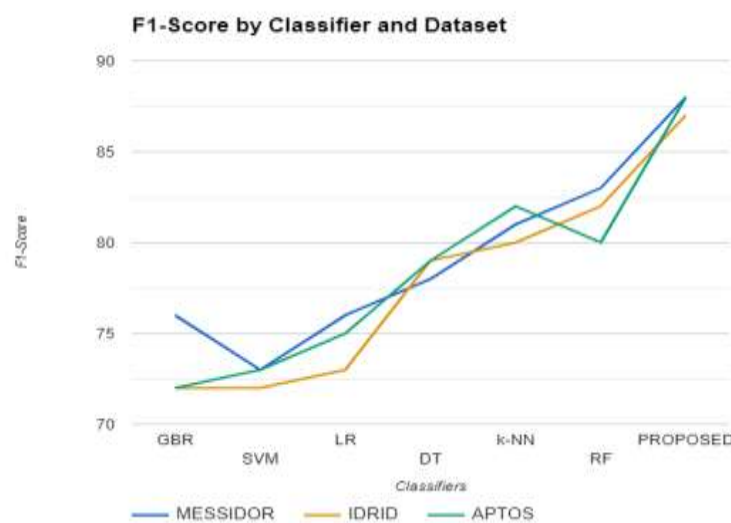


Figure 8: F1-Score Comparison

The proposed technique consistently outperforms other classifiers in terms of precision, recall, and F1-score across all three datasets, demonstrating its effectiveness in various domains. This superior performance suggests that the proposed technique could be a valuable tool for tasks requiring accurate classification in different scenarios. Further investigation into the specific characteristics and inner workings of the proposed technique would be beneficial to understand its strengths and potential limitations.

4. CONCLUSION:

The retina is the inner surface of the human eye and functions as an eye film. Light beams pass through the nerve of the optic, transmitting signals in the form of electricity to the brain. From the eye, the optic nerve connects to the brain. More fructose or glucose will collect in the blood as a result. Because of the improper distribution, these build-up causes affect the vessels of blood and the cells' oxygen levels. Compared to manual diagnoses, automatic DR detection methods are both times- financially advantageous and highly effective. A diagnostic is more prone to errors and demands more work than an automated methodology. A subset of machine learning (ML) approaches called deep learning (DL) uses layers of hierarchical nonlinear processing to classify patterns and learn unsupervised features. Medical image analytics using DL include detection, segmentation, registration, classification, and picture retrieval. Recently, the DR diagnostic process has made extensive use of DL. When numerous diverse sources of heterogeneity are used, it can successfully learn the features of the incoming data. It uses DL-based techniques, including autoencoders, CNN, limited Boltzmann machines, and sparse coding.

5. REFERENCES:

- [1]. R. Bourne, J. D. Steinmetz and S. Flaxman, "GBD 2019 Blindness and Vision Impairment Collaborators; Vision Loss Expert Group of the Global Burden of Disease Study. Trends in prevalence of blindness and distance and near vision impairment over 30 years: an analysis for the Global Burden of Disease Study", *Lancet Glob Health*, vol. 9, no. 2, pp. e130-e143, 2021.
- [2]. Swamy, S. R., Praveen, S. P., Ahmed, S., Srinivasu, P. N., & Alhumam, A. (2023). Multi-features disease analysis based smart diagnosis for covid-19. *Computer Systems Science and Engineering*, 45(1), 869-886.
- [3]. Sirisati, Ranga Swamy, et al. "Cancer Sight: Illuminating the Hidden-Advancing Breast Cancer Detection with Machine Learning-Based Image Processing Techniques." 2023 International Conference on Sustainable Communication Networks and Application (ICSCNA). IEEE, 2023.
- [4]. S Sivaprasad and E. Pearce, "The unmet need for better risk stratification of non-proliferative diabetic retinopathy", *Diabet Med*, vol. 36, no. 4, pp. 424-433, 2019.
- [5]. Kumar, C. Srinivasa, et al. "A Smart Recommendation System for Medicine using Intelligent NLP Techniques." 2022 International Conference on Automation, Computing and Renewable Systems (ICACRS). IEEE, 2022.
- [6]. A. M. Mutawa, Sai Sruthi, "Diabetic Retinopathy Classification using Vision Transformer", 2022 6th European Conference on Electrical Engineering & Computer Science (ELECS), pp.25-30, 2022.
- [7]. Aashdeep Singh, P. Murugeswari, S D Prabu Ragavendiran, Amanpreet Kaur, Gurpreet Singh, Sathiyamoorthy Margabandu, "AI-based Chatbot for Physically Challenged People", 2022 International Conference on Edge Computing and Applications (ICECAA), pp.1039-1044, 2022.
- [8]. SWAMY, S. RANGA, C. SRINIVASA KUMAR, and A. Gauthami Latha. "AN EFFICIENT SKIN CANCER PROGNOSIS STRATEGY USING DEEP LEARNING TECHNIQUES."
- [9]. Anila Sebastian, Omar Elharrouss, Somaya Al-Maadeed, Noor Almaadeed, "A Survey on Deep-Learning-Based Diabetic Retinopathy Classification", *Diagnostics*, vol.13, no.3, pp.345, 2023.

- [10] .Sirisati, Ranga Swamy, et al. "An Enhanced Multi Layer Neural Network to Detect Early Cardiac Arrests." 2021 5th International Conference on Electronics, Communication and Aerospace Technology (ICECA). IEEE, 2021.